

AKHILESH KUMAR MISHRA

PYTHON FULL STACK DEVELOPER | DJANGO · FASTAPI · REST APIS

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PROFILE

500,000+ transactions a day for 2 million+ users on India national e-governance platforms, served by Django REST Framework and async FastAPI services. Took transaction query p95 down 45% by partitioning and indexing PostgreSQL, cleared 35% of the peak backlog onto Celery and Redis workers, and streams live status over WebSockets in under 500ms. Reads the query plan before adding the index.

CORE TECHNOLOGY

Core Python: Python, Django, Django REST Framework, FastAPI, Flask, Celery, SQLAlchemy, Alembic, asyncio

APIs & Security: RESTful APIs, OpenAPI/Swagger, JWT, OAuth 2.0, RBAC, rate limiting, idempotency keys, FastAPI WebSockets, API versioning

Data & Messaging: PostgreSQL, MySQL, Redis, Celery queues, covering indexes, partitioning, query plan analysis

Testing & DevOps: pytest, Docker, Kubernetes, GitHub Actions CI/CD, Linux, Postman, Prometheus, Grafana, React.js, JavaScript (ES6+)

EXPERIENCE

Software Development Engineer | **CSC e-Governance Services India Ltd. — India's national digital services platform, 2M+ users**

Ministry of Electronics & Information Technology, Govt. of India

12/2023 – PRESENT | New Delhi, India

- Scaled AEPS, MATM and VATM orchestration to **500,000+ daily transactions** at a **99% success rate**, by making the **Django REST Framework** payment path idempotent with request keys, backoff retries and nightly reconciliation.
- Took transaction query p95 down **45%** across **PostgreSQL** and **MySQL**, by partitioning the hot tables and adding covering indexes chosen from the query plan.
- Cleared **35%** of the peak-traffic backlog, by moving slow work off the request path onto **Celery** workers with **Redis** broking and idempotent task retries.
- Delivered live transaction status in under **500ms**, by streaming updates over **FastAPI WebSockets** instead of client polling.
- Contained a bad release to one service instead of the whole portal for **2 million+ users**, by moving monolithic web services onto async **FastAPI** and **Django** microservices behind a versioned REST gateway.
- Cut authentication latency **50%**, by issuing **JWT** access/refresh pairs with **RBAC** claims and caching OTP verification state in **Redis**.

Software Engineer | **Contract engagement — field-workforce logistics, 1,000+ agents**

Real-Time Telemetry & Geofencing Platform

06/2023 – 11/2023 | Remote, India

- Built a workforce telemetry and geofencing system for **1,000+ field agents** end to end, on **FastAPI** and async **SQLAlchemy** with **Alembic** migrations.
- Held **99.9% uptime** while absorbing **10,000+ concurrent** location pings, by batching writes through a **Celery** and **Redis** ingestion pipeline.
- Brought dashboard load time down **30%**, by wiring the **React.js** telemetry view to **FastAPI WebSocket** endpoints instead of polling.

SELECTED PROJECTS

DigiPay API Platform

Python / FastAPI / PostgreSQL / Docker / GitHub Actions

FastAPI service with versioned routers, correlation-ID request tracing, auth and rate-limit middleware and a documented deprecation policy, packaged with Docker and a CI pipeline. github.com/GitHub-akhilesh/Django_apis_digipay

Polyglot Task Microservices

Python / Django / Flask / FastAPI / Docker / Kubernetes

Django REST API, Flask utility service and FastAPI notification service running side by side under Docker Compose with Kubernetes manifests, fronted by a React web client. github.com/GitHub-akhilesh/To-do-list-app

Facial Recognition Attendance System

Python / OpenCV / Pandas / Tkinter / MySQL

Automated attendance capture in Python using OpenCV Haar Cascade and LBPH recognition, with MySQL persistence and Pandas-generated Excel reports. github.com/GitHub-akhilesh/automatic-attendance-through-face-detection

Async Telemetry Ingestion & Reporting Service

Python / FastAPI / Celery / Redis / PostgreSQL / Pandas

FastAPI ingestion service with Celery and Redis task queues writing to partitioned PostgreSQL tables, plus Pandas reporting jobs over aggregated telemetry.

EDUCATION

Bachelor of Technology (B.Tech.) | Kanpur Institute of Technology, Kanpur, UP

2019 – 2023

CERTIFICATIONS

IIT Kanpur Cyber Security Certification

ACHIEVEMENTS

- HackerRank 4-Star Gold badge in SQL.